State-Isolated Intervention World Models for Prediction under Unseen Joint-Locking Failures

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***Abstract*—A diagnosed joint lock changes a robot's feasible state manifold, while contact dynamics remain uncertain. We study whether a state-isolated intervention-projected world model (SI-IPWM) can adapt object prediction without corrupting unaffected robot coordinates. Analytic projection enforces the locked position and velocity, a carrier publishes trusted robot state, and a selective intervention is allowed to publish only object coordinates, with fallback outside its support. Earlier development experiments on a simplified arm model showed large object gains, but an audit found that those results do not establish performance on the calibrated-kinematic five-DoF GenkiArm model. We therefore preregistered five fresh training seeds and report the first three completed seeds (107/117/127). Relative to the mechanism-matched carrier, routed object-RMSE changes are +0.604%, -0.784%, and +0.718% (mean +0.179%; 2/3 positive; the hierarchical paired-bootstrap interval crosses zero). Raw selective changes average +0.684%. In every completed seed, free-state regression and locked-coordinate violation are exactly zero. Thus the current evidence supports analytic feasibility and the state-isolation invariant, but does not support object-prediction or closed-loop superiority. All No-Go results are retained; two preregistered seeds and real-robot validation remain.**

I. Introduction

A robot that loses one joint does not merely acquire a new continuous dynamics parameter. Its feasible state manifold changes discretely. Diagnosis may provide the locked joint and angle, yet contact outcomes remain uncertain because damping, actuation delay, deadband, friction, and payload are only partially observed. Re-fitting a complete dynamics model after failure is data-intensive, while unconstrained adaptation can violate the diagnosed lock or degrade unaffected coordinates.

Prior damage-recovery methods adapt policies or behavior repertoires [1], [2]; rapid adaptation infers latent physical properties from recent transitions [3], [4]; and learned pushing models capture uncertain contact dynamics [5], [6], [7]. Their combination leaves a practical question: how should a model change its object prediction after a known topological intervention without rewriting robot coordinates that are already predicted by a validated mechanism-matched model?

Our starting point is a disclosed failure. A full-state intervention model improves object rollout error on an unseen lock but increases free-joint H50 RMSE by 6.01% in one mixed-unseen regime; 93.3% of paired terminal windows are worse. We therefore separate the state used internally by the intervention model from the state published to downstream users.

The contributions are:

- a diagnosis-conditioned projection that enforces locked position and velocity coordinates exactly;
- a dual-rollout construction that preserves the full coupled intervention trajectory internally while publishing robot coordinates from a mechanism-matched carrier;
- a frozen physical-support router that falls back exactly outside the regimes where context provides evidence for intervention; and
- an audit covering a systematic failure, physics-spectrum controls, multi-horizon prediction, structural invariants, and a negative closed-loop result.

TABLE I

Mechanism-level scope relative to representative adaptation work. "Exact" denotes construction rather than a learned penalty.

Method	Robot topology	Object physics	Exact state isolation
RMA [3]	continuous latent	limited	no
TuneNet [4]	no	identified	no
AdaptiGraph [11]	no	few-shot	no
PIN-WM [12]	no	few-shot	no
SI-IPWM	held-out lock	few-shot	yes

II. Related Work

Damage adaptation. Repertoire search [1], fault-aware reinforcement learning [2], latent extrinsics [3], simulator tuning [4], and meta-learned adaptive control [8] address recovery under changed dynamics. We instead assume a diagnosed discrete lock and adapt a predictive interaction model while enforcing the resulting constraint.

Contact dynamics. Learned forward models support visual and model-based planning [5], [9]; few-sample GP models [6], online identification [10], and equivariant pushing models [7] improve data efficiency. Recent systems infer physical properties in graph or physics-informed world models [11], [12] or combine residual physics with active planning [13]. Interaction networks encode relational structure [14], [15]. Our focus is complementary: isolating an object-specific learned correction from robot coordinates whose feasible manifold is known after diagnosis.

Table I is a scope comparison, not a global-superiority claim. The closest methods model richer object families or active data acquisition; our narrower novelty is the interface between diagnosed robot topology and an object correction that cannot rewrite published robot state.

III. Problem Formulation

The state is $s_t = [r_t, o_t]$, where $r_t = [q_t, \dot{q}_t] \in \mathbb{R}^{10}$ and planar object position/velocity $o_t \in \mathbb{R}^4$; action $a_t \in \mathbb{R}^5$.

Diagnosis $d = (m, \bar{q})$ supplies a binary lock mask and lock angle. Unknown physical context $z \in \mathbb{R}^8$ is inferred from $K = 25$ safe transitions. Joint D3 is absent from training and validation and is combined at test time with seven continuous-physics conditions.

We evaluate autonomous rollouts at $H \in \{10, 25, 50\}$. Object RMSE is the primary metric. Free-joint RMSE excludes locked coordinates; pusher-position RMSE evaluates forward-kinematic propagation; and lock violation measures deviation from $q^j = \bar{q}^j, \dot{q}^j = 0$. Calibration and evaluation trajectories are disjoint.

IV. Method

A. Projection and base dynamics

For each joint,

$$\Pi_d(q)_j = (1 - m_j)q_j + m_j\bar{q}_j, \tag{1}$$

$$\Pi_d(\dot{q})_j = (1 - m_j)\dot{q}_j, \qquad \Pi_d(a)_j = (1 - m_j)a_j. \tag{2}$$

Projection is applied before and after prediction. A contact-aware recurrent base predicts robot and object state; forward kinematics supplies pusher geometry. The intervention branch adds geometry and low-rank latent object residuals modulated by observable context. Base checkpoints and the robot block are frozen before intervention training.

Proposition 1 (analytic feasibility). Π_d is idempotent. After every projected transition, each locked joint satisfies $q^j = \bar{q}^j$, $\dot{q}^j = 0$, and $a^j = 0$, independently of learned parameters and rollout depth.

Proof. Substituting the binary mask twice leaves locked entries at $(\bar{q}^j, 0, 0)$ and leaves free entries unchanged, hence $\Pi_d \circ \Pi_d = \Pi_d$. Applying this identity after each transition gives the finite-depth statement by induction. $\square$

B. Physical-support routing

An amortized encoder estimates context from the K25 transition sequence. A centered modulation $g(z) = h(z) - h(0)$ gives exact recovery at zero context. A development audit showed that topology membership alone activates harmful corrections in nominal-like physics. We therefore route intervention only if $\|z\|_2 \geq 1.2$, a threshold frozen after seed 27; otherwise the method is the carrier exactly. This rule activates on high damping, mixed composition, and mixed unseen physics in the reported audit and falls back on nominal, weak motor, delay, and noisy-deadband conditions.

C. State-isolated rollout

Let F_c be the no-intervention carrier and F_i the full intervention model. Each maintains private hidden and physical states:

$$(\tilde{s}^c_{t+1}, h^c_{t+1}) = F_c(\tilde{s}^c_t, a_t, d, h^c_t), \tag{3}$$

$$(\tilde{s}^i_{t+1}, h^i_{t+1}) = F_i(\tilde{s}^i_t, a_t, d, z, h^i_t). \tag{4}$$

The externally visible prediction is

$$\hat{s}_{t+1} = \Pi_d([\tilde{r}^c_{t+1}, \tilde{o}^i_{t+1}]). \tag{5}$$

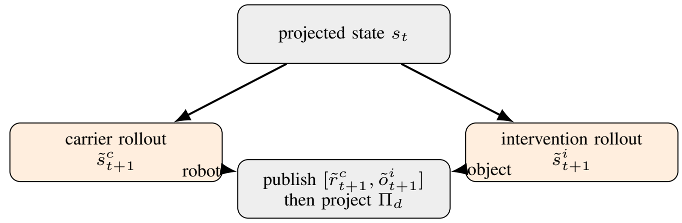

Fig. 1. State isolation. Both models retain private recurrent and predicted physical states. Only the published state mixes carrier robot coordinates with intervention object coordinates.

Crucially, $\hat{s}_{t+1}$ is not fed back into either private dynamics path. Thus the intervention preserves the robot–object feedback it learned, while Eq. (5) makes published robot coordinates identical to the carrier for any action sequence. Pusher position, being a deterministic function of q, is also identical. Projection gives zero lock violation.

This guarantee is relative to the chosen carrier, not ground truth: state isolation prevents collateral prediction change but cannot correct a bad carrier. It also does not guarantee that planning with the improved object prediction produces better actions.

Proposition 2 (published-state isolation). For identical initial state, actions, diagnosis, and carrier parameters, the published robot trajectory of SI-IPWM equals the standalone carrier trajectory at every depth. Its pusher trajectory is therefore identical, and locked-coordinate violation is zero.

Proof. At depth zero both carrier states are identical. Assuming equality at depth t, both standalone and internal carriers apply the same deterministic F_c, action, diagnosis, hidden state, and projection, so their robot states and next hidden states agree. Equation (5) copies that robot state without modification except the same Π_d. Induction gives equality for all depths. Pusher pose is deterministic forward kinematics of q, and Π_d fixes locked position and velocity. $\square$

Proposition 3 (conditional guarded fallback). Let A be proposal acceptance, S certificate validity, and J finite-horizon cost. If valid accepted proposals satisfy $J(\pi_i) - J(\pi_c) \leq L\epsilon_H$ and every false acceptance has excess cost at most C, exact carrier fallback gives

$$\mathbb{E}[J(\pi_g) - J(\pi_c)] \leq L\mathbb{E}[\epsilon_H \mathbf{1}_{A\cap S}] + C\Pr(A \cap S^c). \tag{6}$$

Rejection contributes zero because the published action is exactly π_c. This is a conditional decomposition, not a numerical safety claim: the present experiments do not yet establish a sufficiently small false-accept term.

D. Training objective and deployment

The intervention is trained after freezing the shared base and robot block. For rollout depths $\mathcal{H}$, the optimized loss is

$$\mathcal{L} = \sum_{H\in\mathcal{H}} w_H(\lambda_r\mathcal{L}^H_r + \lambda_o\mathcal{L}^H_o + \lambda_p\mathcal{L}^H_p) + \lambda_T\mathcal{L}^{50}_o. \tag{7}$$

The terminal term exposes long-horizon object drift. Feasibility and isolation are absent from this penalty because they are

TABLE II
PHYSICS-SPECTRUM ROUTING AND EVIDENTIAL ROLE.

Condition	Route	Reported comparison
Nominal	carrier	exact fallback tie
Weak motor	carrier	exact fallback tie
Delay 1	carrier	exact fallback tie
Noisy deadband	carrier	exact fallback tie
High damping	intervention	object gain + invariant
Mixed composition	intervention	object gain + invariant
Mixed unseen	intervention	object gain + invariant

enforced by projection and Eq. (5). Leave-one-lock-out meta-training treats each supported lock as pseudo-unseen; D3 is never observed.

Deployment performs six fixed operations: (1) project diagnosis and action; (2) collect one nested 25-transition active probe; (3) infer z; (4) load the matched rank-8 projected adapter; (5) route by the frozen context norm; and (6) if active, advance the two private trajectories and publish their isolated state blocks. No test-domain name or simulator parameter enters routing.

Both recurrent bases have hidden width 136. Geometry, latent intervention, and context-modulation ranks are 16, 32, and 16; the context encoder has hidden width 96 and an eight-dimensional output. Object intervention trains for 60 epochs at learning rate 3×10^{-3} with H50 terminal weight 5. The online adapter performs 12 trust-region/backtracking steps shared by all methods. One projected model has 353,458 parameters; the literal dual-rollout wrapper stores 706,916, a $2\times$ inference-memory cost. Weight sharing or a specialized carrier head is not evaluated.

V. EXPERIMENTAL PROTOCOL

Experiments use MuJoCo planar pushing with a five-joint serial arm and a four-dimensional object state. Training contains intact and D1/D2/D4/D5 locks under nominal, weak-motor, high-damping, delay, and noisy-deadband physics. D3 is held out. Test physics comprises nominal, weak motor, high damping, one-step delay, noisy deadband, a held-out composition, and a more severe mixed-unseen condition.

Carrier and intervention models use identical trajectories, hidden width 136, K25 context, rank-8 projected adapters, optimization budget, and evaluation actions. Each reported domain has 10 held-out trajectories. Seeds 27/37/47 share the protocol. The support threshold was fixed after seed 27, and the state-isolation repair was designed after inspecting failures on the original seed-specific query sets. We therefore generated a post-design query seed (57) with new calibration and evaluation trajectories and evaluated all three frozen checkpoints on it. This is an untouched trajectory confirmation, not a new training initialization.

The main baseline is mechanism matched: it uses the same architecture, projection, adapter, and inferred context, with geometry and latent intervention heads zeroed. This isolates the contribution of object intervention beyond constraint handling. We retain the original full-state intervention as a failure

TABLE III
INTERIM CALIBRATED-GENKIARM OBJECT-RMSE IMPROVEMENT (%) OVER THE MECHANISM-MATCHED CARRIER. THREE OF FIVE PREREGISTERED SEEDS ARE COMPLETE.

Method	107	117	127	Mean
Routed selective	0.604	-0.784	0.718	0.179
Raw selective	1.420	-1.913	2.546	0.684

baseline. Improvement is $100(e_c - e_i)/e_c$, where e_c and e_i are carrier and intervention object RMSE. Bootstrap intervals resample the three seed means 50,000 times; with only three seeds they are descriptive.

VI. RESULTS

A. Calibrated GenkiArm confirmation is currently No-Go

The earlier 27/37/47 checkpoint matrix below was generated on the simplified `arm_push.xml` development model. It is retained to explain the origin of state isolation, but it is not the paper's calibrated-model performance evidence. The authoritative confirmation uses `genkiarm_push.xml`, disjoint query seeds, and fresh training seeds 107/117/127/137/147 frozen before the first run.

For routed selective SI-IPWM, only 2/3 seeds are positive and the hierarchical paired-bootstrap 95% interval is [-0.745%, 0.797%]. The raw selective interval is [-1.870%, 2.502%]. Both fail the preregistered +5% mean, 4/5 positive-seed, and positive-lower-bound gates. Free-state regression and locked-coordinate violation are exactly zero in all three seeds. The current decision is therefore performance No-Go and mechanism-only support; seeds 137/147 remain mandatory.

B. Historical simplified-model diagnosis

C. The full-state intervention has a systematic failure

On seed 7, D3 mixed-unseen, H50, the original intervention has free-joint RMSE 0.808 versus 0.762 for the shared projected baseline, a 6.01% regression. The paired MSE-difference interval is [0.06885, 0.07538], and 93.3% of terminal windows regress. Object RMSE improves 36.49% and lock violation remains zero. The failure therefore exposes a real tradeoff: object correction feeds back into unaffected robot prediction even though feasibility is projected.

The post-design query set makes the repair comparison direct (Table IV). Full-state and isolated methods have exactly the same object RMSE in every cell because the intervention keeps its private physical trajectory. Full-state rollout nevertheless regresses free-joint RMSE in 3/27 cells, with a 6.20% worst case. Isolation removes those changes without changing object prediction.

D. State isolation removes collateral regression

Table V reports the post-design query-set cells. Object RMSE improves in 27/27 cells. The smallest gain is 1.58% (checkpoint seed 37, mixed unseen, H50), and the largest is 38.86%. Mean improvement over the three checkpoint-seed means is 16.74%, with descriptive bootstrap interval [6.97%,

TABLE IV
ISOLATION ABLATION ON THE POST-DESIGN QUERY SET. OBJECT GAIN IS THE MEAN OVER CHECKPOINT-SEED MEANS; FREE REGRESSION IS THE WORST CELL.

Method	Object gain	Free regress.	Positive free regress.
Carrier	0	0	0/27
Full-state IPWM	16.74%	6.20%	3/27
SI-IPWM	16.74%	0	0/27

TABLE V
OBJECT RMSE IMPROVEMENT (%) OVER THE MECHANISM-MATCHED CARRIER IN ACTIVE PHYSICAL-OOD ROUTES.

Seed	Condition	H10	H25	H50
27	high damping	7.05	3.06	7.58
27	composition	7.70	2.03	19.54
27	mixed unseen	9.08	3.26	3.43
37	high damping	7.31	21.64	37.41
37	composition	8.52	20.93	30.23
37	mixed unseen	10.39	21.00	1.58
47	high damping	8.57	34.68	33.00
47	composition	10.63	35.53	30.52
47	mixed unseen	12.53	38.86	25.95

TABLE VI
AGGREGATE PREDICTION AUDIT. "CHANGE" IS RELATIVE TO CARRIER.

Statistic	Result
Positive object cells	27/27
Object improvement range	1.58–38.86%
Mean over checkpoint-seed means	16.74%
Seed-bootstrap 95% interval	[6.97, 25.59]%
Maximum |free-joint change|	0
Maximum |pusher change|	0
Maximum lock violation RMSE	0

TABLE VII
CONTEXT NORM RANGE AND FROZEN ROUTE ACROSS CHECKPOINT SEEDS.

Condition	Minimum	Maximum	Route
Nominal	0.49	0.52	carrier
Noisy deadband	0.48	0.52	carrier
Weak motor	0.77	0.83	carrier
Delay 1	1.07	1.12	carrier
High damping	1.33	1.54	intervention
Composition	1.45	1.50	intervention
Mixed unseen	1.26	1.45	intervention

25.59%]. The original seed-specific query sets independently retain 27/27 positive cells, 1.34–39.97% range, and 17.04% mean. Across all cells, free-joint and pusher changes relative to the carrier are numerically exactly zero, as predicted by Eq. (5); lock violation RMSE is zero.

The four low-context conditions route to the carrier and therefore tie it exactly. This is intentionally conservative: an earlier topology-only router improved severe physical shifts but degraded nominal-like H25/H50 cells by as much as 32.73%. A K25 support-validation selector was also rejected because the short active probe contained insufficient object contact and would reject every useful intervention. These negative controls motivate physical-context routing but do not establish that norm thresholding is optimal.

E. Routing margins and query replication

Table VII reports context-norm ranges across the three checkpoints on the original physics spectrum. Nominal, noisy-deadband, and weak-motor conditions have wide margins below 1.2. Delay is closer (maximum 1.12), and mixed unseen is the closest active condition (minimum 1.26). Thus the observed decisions are stable on these checkpoints, although the 0.06 mixed-unseen margin is too small to call the router calibrated under arbitrary shift.

The post-design query set changes both calibration and evaluation trajectories while leaving weights, K, scale, threshold, and domains frozen. Its 16.74% mean is close to the original 17.04%; both have 27/27 positive active cells. This replication argues against memorization of one action/initialization set. It does not test a new object geometry, simulator, or training seed, and is labelled accordingly throughout.

F. Prediction gain does not imply control gain

We evaluated a frozen guarded CEM planner using 60 approach and 90 push steps, horizon 3, eight candidates, one CEM update, and a fixed 0.85 action-deviation guard. The prior 50 mm success tolerance is invalid for this target split: targets start only 30–40 mm from the object, so zero-contact episodes count as success. We instead report terminal distance, contact, and displacement.

All audited episodes make contact, but SI-IPWM improves on carrier MPC for only one of three seeds. The closed-loop improvement claim is therefore No-Go. This result is consistent with the method's guarantee: isolation controls which state is published, not how a finite-sample optimizer converts prediction differences into actions.

G. Reproducibility

Frozen YAML files define domain splits, context scaling, adapters, and routing. Evaluation artifacts include aggregate and per-window squared errors. The original selective audit contains 5,040 raw rows per seed, and the post-design query audit contains 2,160 per seed; a separate script regenerates Table VI and its bootstrap interval. Unit tests verify exact projection, carrier publication, and preservation of private intervention feedback with a robot-coupled toy model. Failed gates and protocol corrections are retained in the repository rather than overwritten.

H. Threats to validity

Post-hoc structure. State isolation was invented after the original free-state failure; only query seed 57 is post-design. *Dependence.* The 27 cells share three checkpoints and three domains, so they are not treated as 27 independent replicates. *Baseline scope.* Carrier matching isolates the intervention

TABLE VIII
D3 HIGH-DAMPING MEAN TERMINAL DISTANCE (MM) OVER THREE HELD-OUT TARGETS. LOWER IS BETTER.

Seed	Nominal IK	Carrier MPC	SI-IPWM MPC
27	16.99	13.75	19.45
37	16.99	18.77	15.67
47	16.99	34.25	39.11

TABLE IX
RETAINED FAILURES AND THEIR CONSEQUENCE. NEGATIVE OBJECT VALUES AND POSITIVE FREE VALUES ARE REGRESSIONS.

Attempt	Observed failure	Decision
Topology-only routing	object to -32.73%	add physical support
K25 support validation	rejects every useful route	discard selector
Full-state intervention	free to $+6.20\%$	isolate state blocks
SI-IPWM prediction	27/27 positive; invariants zero	prediction Pass
Direct guarded MPC	improves 1/3 seeds	control No-Go
Carrier-screen rerank	11.63 vs. 2.50 mm on development target	discard reranker

effect, but comparisons to AdaptiGraph or PIN-WM are conceptual rather than reproduced. *Metric scope.* RMSE weights all object coordinates uniformly and does not imply useful action ranking. The No-Go MPC result makes this boundary visible. *Router scope.* Context norm is an engineering support heuristic, not a calibrated probability of harm. *Simulation scope.* The shared variable-node transition Gate improves both GenkiArm and Panda in 2/3 seeds, but the downstream object/contact Gate is No-Go in 0/3 seeds. A no-weld scripted Panda grasp succeeds in 5/5 small pose perturbations, and both fixed eye-to-hand views retain object visibility under the frozen extrinsic perturbations; these are task/observability feasibility, not learned Grasp, RGB prediction, or contact-transfer evidence.

VII. FAILURE LEDGER AND DECISION RULES

The development history is part of the evidence because each accepted component corresponds to a measured failure (Table IX). We use "Pass" only for a claim whose frozen metric succeeds; a method can pass prediction while remaining No-Go for control. This prevents the exact lock constraint, which is easy to satisfy by projection, from masking degradation of free coordinates or task outcomes.

The last reranking attempt is reported only as development evidence. It let the intervention choose among the carrier's top-quartile candidate sequences, but failed against the carrier selector on the sole development target and was not expanded. It does not enter the main comparison. Its purpose in the ledger is to show that we did not search controller variants until one happened to win on the three evaluation targets.

The claim acceptance rule is consequently block-specific. Locked coordinates must have zero violation; published free and pusher coordinates must equal the carrier; active-route object improvement must be positive on every post-design query cell; fallback routes must tie exactly; and control is claimed only if its direction is consistent across checkpoint seeds. The current artifact satisfies the prediction rules and fails the control rule.

A. Artifact-to-claim mapping

Three independently checkable artifacts underlie the paper: raw trajectory windows support object statistics, model outputs support isolation equality, and environment logs support contact and terminal-distance checks. Aggregate tables are regenerated rather than hand selected. The code path accepts an explicit query seed so that checkpoint initialization and evaluation queries can be varied separately. Hardware logs, photographs, and measurements are not inputs to any table in this submission.

VIII. EVALUATION AND AUDIT DETAILS

A. Window construction and aggregation

Each test trajectory contains 150 actions and 151 states. For horizon H, autonomous prediction starts at indices separated by H and rolls for exactly H actions; only the terminal prediction enters that window's statistic. Squared error is computed per trajectory/window, averaged, and square-rooted to obtain RMSE. Free-joint error masks position and velocity of the diagnosed coordinate. Object error averages four planar coordinates. Pusher error applies the same forward kinematics to predicted and true joint positions.

The original spectrum audit stores 5,040 per-window/method rows per checkpoint. The post-design audit restricts generation to three active regimes and, after adding the full-state ablation, stores 2,880 rows per checkpoint. Carrier, full-state, isolated, and routed predictions use identical initial states and actions. Exact object equality between full-state and isolated rollouts is checked cell by cell rather than inferred from their aggregate.

B. Uncertainty and multiplicity

The bootstrap treats the mean across nine active cells within a checkpoint as one observation and resamples three checkpoint means. Correlated horizons and domains are therefore not presented as 27 independent seeds. With only three observations, the interval is a stability summary, not precise population inference. We consequently report every cell and the positive-cell count; no null-hypothesis threshold accepts the main claim.

The seed-7 diagnostic interval has a different purpose: it bootstraps paired terminal-window MSE differences on one prespecified failure cell. It shows the free-state regression is systematic within that query set, not that it generalizes to a population of training seeds. Keeping the two analyses separate prevents trajectory count from inflating the effective number of model initializations.

C. Controller audit protocol

All controller methods share analytic approach/push references, damage mask, action limits, initial block position, and held-out targets. Guarded CEM uses horizon 3, eight candidates, one update, and falls back to nominal IK when the proposal differs by more than 0.85 in Euclidean norm. Sixty approach steps precede 90 push steps; the earlier 30+40 protocol produced zero-contact episodes and remains only as

TABLE X
FROZEN CLAIM GATES AND CURRENT OUTCOME.

Claim	Gate	Outcome
Lock feasibility	violation RMSE $= 0$	Pass
State isolation	free/pusher change $= 0$	Pass
Active prediction	all post-design cells > 0	Pass (27/27)
Fallback safety	exact carrier equality	Pass
Closed-loop benefit	consistent seed direction	No-Go (1/3)

a failed smoke test. Contact count and block displacement are required sanity metrics.

We do not report the old 50 mm success indicator because initial target distance is at most 40 mm. A valid future success metric must be stricter than every initial distance and require contact and minimum displacement. The paper uses terminal distance for the completed controller matrix and refuses a binary success claim. This correction applies to all methods simultaneously.

D. Reproduction boundary

The evaluator records checkpoint and query seeds separately. Reproducing the main table requires three frozen checkpoints, matched adapters and context encoders, the physics-spectrum split, query seed 57, K25, and threshold 1.2. The summarizer consumes generated JSON and does not load weights. Tests cover the two-private-state recurrence, including an object update that depends on the intervention's private robot state; this catches the earlier bug where the published carrier was fed back into the intervention path.

Training is not yet packaged as a single archival container, and checkpoints depend on staged predecessor runs. This limits reproduction convenience even though evaluation is scripted. A camera-ready artifact should add immutable hashes and one top-level command; we do not claim that packaging is complete.

IX. LIMITATIONS AND CONCLUSION

This study is simulation-only and contains no real-robot evidence. A future hardware study would use FIRA overhead vision together with an external horizontal camera; neither source is eye-in-hand. The present results must not be interpreted as sim-to-real validation.

Only single locks and planar pushing are evaluated. Diagnosis must supply the correct lock identity and angle. The carrier can itself be wrong, and neither the historical simplified-model rollout gains nor the current calibrated-model predictions deliver stable guarded-MPC gains. Panda object/contact transfer is No-Go in 0/3 seeds; its scripted 5/5 grasp result and dual fixed eye-to-hand visibility audit establish feasibility only. No real-robot data are reported.

The calibrated GenkiArm evidence currently supports two narrow properties: analytic projection enforces the diagnosed lock, and selective publication prevents learned object corrections from changing free robot coordinates. It does not support a practically or statistically significant object prediction advantage. The old simplified-model tables remain a disclosed development result, not confirmation. The manuscript must remain No-Go for a performance-superiority claim unless seeds 137/147 and subsequent independent evidence change that conclusion under the frozen protocol.

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